

Pricing Off Accounting Costs

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Preliminary—subject to change!

Pricing and Inventory Costs: Fed Beige Book

- “Several contacts noted they were still working through pre-tariff inventories, thus delaying price adjustments.”

— Federal Reserve Bank of Atlanta, June 2025

- “Several contacts across different retail sectors phased in price increases as old inventories were sold and replaced with goods subject to tariffs. Retailers expect additional price increases as this phase-in continues.”

— Federal Reserve Bank of San Francisco, Sep 2025

- “A coffee roaster noted that while tariffs on coffee have largely been lifted, selling **prices will only go down** once the stock of inventory acquired at higher costs has been cleared.”

— Federal Reserve Bank of New York, Jan 2026

Pricing and Inventory Costs: Earnings Calls

- **Manufacturers:** “We have the opportunity for those duties that we pay, those tariffs that we pay, to recover that when we deliver the aircraft. Now, you know, it creates a little bit of a cash flow timing issue that we’ll have to work through.”

— Boeing, Apr 23 2025

- **Distributors:** “We anticipate implementing additional pricing actions as inventory affected by tariffs moves through our cost of sales.”

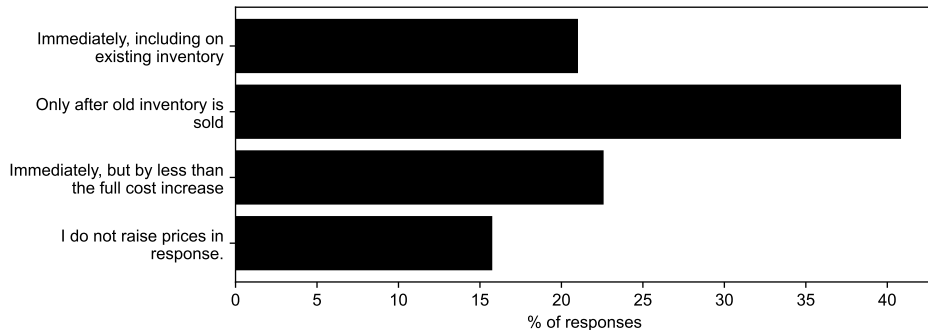
— Global Industrial, July 29 2025

- **Retailers:** “Our pricing algorithms seek to maintain a balance between factoring in our higher wholesale costs and keeping us market priced. In practice, this means that for a short period of time, inventory-carrying retailers may have a slight cost advantage. This should dissipate over the next couple of quarters as the market rebalances, pre-tariff inventory is depleted, and all retailers incur higher wholesale costs reflective of tariffs.”

— Wayfair, Oct 31 2019

Pricing and Inventory Costs: Surveys

- “Suppose your input cost rises today, but you still have inventory bought at lower cost. When do you raise your sales price?”



Note: Survey of 400 firm price-setters by Schoenle (2026).

Pricing and Inventory Costs

- Standard theory: price off marginal cost = current replacement cost.
- Historical purchase cost is sunk and thus irrelevant!
- Part of broader set of puzzles:
 - Firms' prices reflect hedging gains/losses. [Evidence from airlines in paper]
 - Firms report passing through changes in overhead costs.
 - Firms' prices do not respond much to anticipated future costs, demand shocks.

Two Central Propositions

- ① Firms use backward-looking cost measures (“accounting costs”).
 - “The need for objective and systematic measurement forces the cost standard to be backward-looking—or at least sideward- rather than forward-looking.”
— Okun (1981)
- ② Firms largely set prices using these cost measures, rather than replacement costs.
 - “The reliance of firms on historical costs is evident in many empirical studies.... It is reasonably predictable that today’s changes in cattle prices will show up on the supermarket counter only after a lag; the butcher does not keep changing price tags in pace with changes in some putative replacement cost.”
— Okun (1981)

Roadmap

- ① Review of how accounting costs calculated.
- ② Measure extent of “accounting cost pricing” using public firm 10Qs.
 - 60–80% weight on accounting costs.
 - Replacement cost pricing largely in industries with auction exchanges / reference prices.

Roadmap

- ① Review of how accounting costs calculated.
- ② Measure extent of “accounting cost pricing” using public firm 10Qs.
 - 60–80% weight on accounting costs.
 - Replacement cost pricing largely in industries with auction exchanges / reference prices.
- ③ Principal–agent model of pricing off accounting costs.
 - Ingredient #1: Replacement costs unverifiable, but accounting is verifiable / contractible.
 - Ingredient #2: Without monitoring, risk of manager discretion.
 - Mechanism can explain other pricing puzzles.

Roadmap

- ④ Price dynamics resemble sticky-price models.
 - But delays due to production lags / inventory stocks, not length of price spells.
 - Anticipation can delay pass-through (stocking up), opposite sticky-price models.
 - Measured markups can be acyclical/procyclical even if true markup countercyclical.
- ⑤ Quantitative estimates for U.S.
 - Price changes for goods-producing industries take avg. 3–4 months to show up in consumer prices.
 - Delays in tariff pass-through consistent with evidence.

Selected Related Literature

- **Cost-based pricing:** Gordon (1981), Blinder et al. (1998).
 - **Firms inattentive to future costs:** Gordon (1990), Mankiw and Reis (2002), Woodford (2003), Minton and Wheaton (2022), Afrouzi et al. (2024).
⇒ Anticipatory forward buying, and firms price low even while importing at higher price!
 - Okun (1981), Gordon (1981): “Many firms, however, appear to base prices on actual costs paid in the past, so-called ‘FIFO’ (first-in-first-out) pricing practices.”
- **Other pricing puzzles:**
 - **Full-cost pricing:** E.g., Hall and Hitch (1939), Altomonte et al. (2015), Bewley (2025).
 - **Response to cost vs. demand shocks:** Cagan (1979), Bils and Chang (2000), Fabiani et al. (2006), Gagnon and Lopez-Salido (2020), Kohler et al. (2026).
- **Verifiable info and firm monitoring:** E.g., Meckling and Jensen (1976), Holmström (1979), Graham et al. (2005), Rogerson (2008, 2011).
- **Fairness:** Kahneman et al. (1986), Rotemberg (2005, 2011), Eyster et al. (2021).

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Estimating the Extent of Accounting Cost Pricing

Principal–Agent Model

Price Dynamics

Macro Rigidity

Review of accrual accounting (FASB and IFRS)

- **Matching principle:** Match timing of expenses to revenues.
- **Cost principle:** Value expenses at historical cost.

Expense Recognition (Matching) Principle

The expense recognition principle (also referred to as the matching principle) states that we must match expenses with associated revenues in the period in which the revenues were earned. A mismatch in expenses and revenues could be an understated net income in one period with an overstated net income in another period. There would be no reliability in statements if expenses were recorded separately from the revenues generated.

For example, if Lynn earned printing revenue in April, then any associated expenses to the revenue generation (such as paying an employee) should be recorded on the same income statement. The employee worked for Lynn in April, helping her earn revenue in April, so Lynn must match the expense with the revenue by showing both on the April income statement.

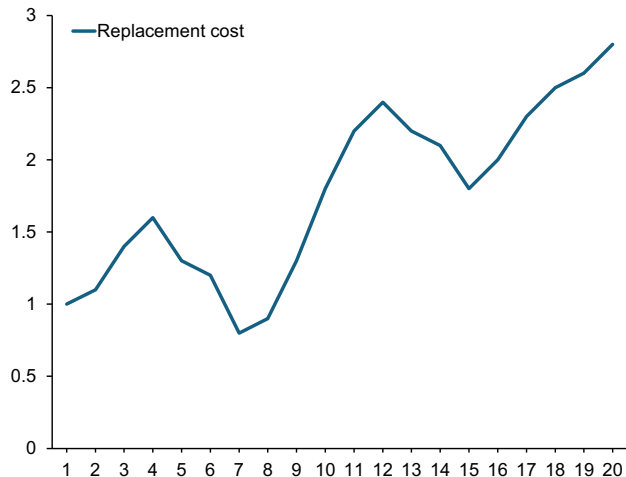
Cost Principle

The cost principle, also known as the historical cost principle, states that virtually everything the company owns or controls (assets) must be recorded at its value at the date of acquisition. For most assets, this value is easy to determine as it is the price agreed to when buying the asset from the vendor. There are some exceptions to this rule, but always apply the cost principle unless FASB has specifically stated that a different valuation method should be used in a given circumstance.

The primary exceptions to this historical cost treatment, at this time, are financial instruments, such as stocks and bonds, which might be recorded at their fair market value. This is called mark-to-market accounting or fair value accounting and is more advanced than the general basic concepts underlying the introduction to basic accounting concepts; therefore, it is addressed in more advanced accounting courses.

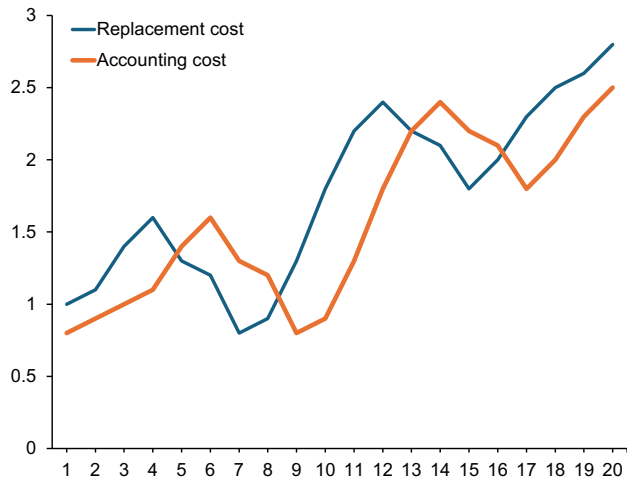
Once an asset is recorded on the books, the value of that asset must remain at its historical cost, even if its value in the market changes. For example, Lynn Sanders purchases a piece of equipment for \$40,000. She believes this is a bargain and perceives the value to be more at \$60,000 in the current market. Even though Lynn feels the equipment is worth \$60,000, she may only record the cost she paid for the equipment of \$40,000.

Example: Replacement vs. accounting costs



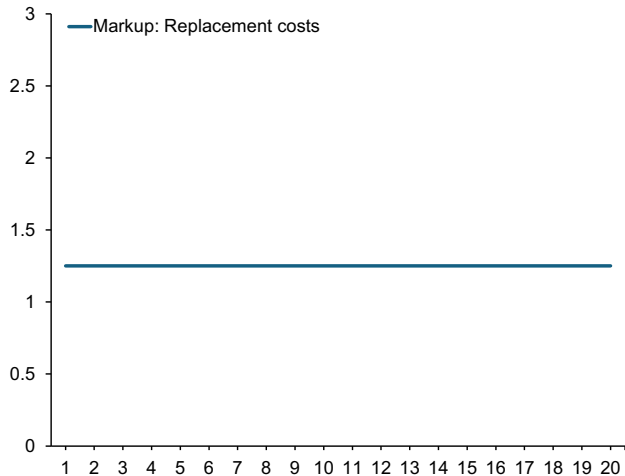
- Replacement costs c_t .
- Inventory stock I_t , units sold D_t .
- Example: D and I constant,
 $\delta = D/I = 0.5$.
(Two months of inventory.)

Example: Replacement vs. accounting costs



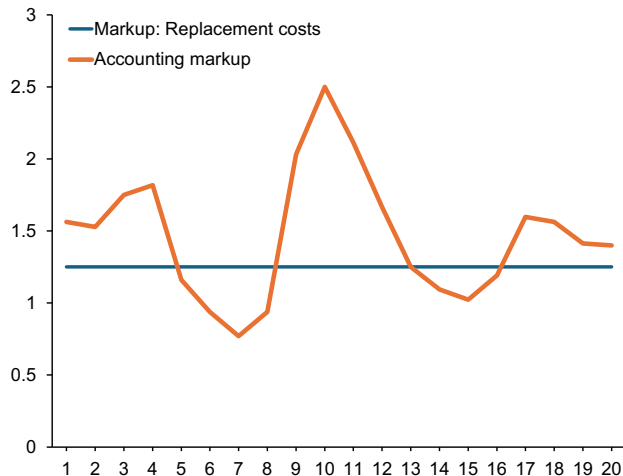
- Replacement costs c_t .
- Inventory stock I_t , units sold D_t .
- Example: D and I constant, $\delta = D/I = 0.5$.
(Two months of inventory.)
- First-in-first-out (FIFO) accounting.
- Cost of goods sold (COGS) $= h_t D_t$,
where $h_t = c_{t-2}$.

Example: Accounting markups



- Suppose optimal price $p_t = 1.25 \times c_t$.

Example: Accounting markups



- Suppose optimal price $p_t = 1.25 \times c_t$.
- Sales / COGS = $1.25 \times c_t / h_t$.
- What would you infer if always accounting markup = 1.25?

Can we estimate the extent of accounting cost pricing?

- Suppose firm sets price = markup $\times$ cost. Denote weight on replacement costs by ϕ .

$$p_t = \mu (\phi c_t + (1 - \phi) h_t).$$

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- Accounting markup (sales / cost of goods sold) is:

$$\frac{\text{Sales}_t}{\text{COGS}_t} = \frac{\mu (\phi c_t + (1 - \phi) h_t) D_t}{h_t D_t} = \mu \left(\phi \frac{c_t}{h_t} + (1 - \phi) \right).$$

- $\phi > 0 \Rightarrow$ prices high relative to accounting cost when $c_t > h_t$.

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- $\phi > 0 \Rightarrow$ prices high relative to accounting cost when $c_t > h_t$.
- How can we tell when $c_t > h_t$?

Can we estimate the extent of accounting cost pricing?

- Key insight: Purchases capitalized into inventory.

$$\text{BookValueInventory}_t = \text{BookValueInventory}_{t-1} - \underbrace{\text{COGS}_t}_{h_t D_t} + \underbrace{\text{Purchases}_t}_{c_t Y_t}.$$

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- Thus,

$$\frac{\text{Purchases}_t}{\text{COGS}_t} = \frac{\Delta \text{BookValueInventory}_t + \text{COGS}_t}{\text{COGS}_t} = \frac{c_t Y_t}{h_t D_t}.$$

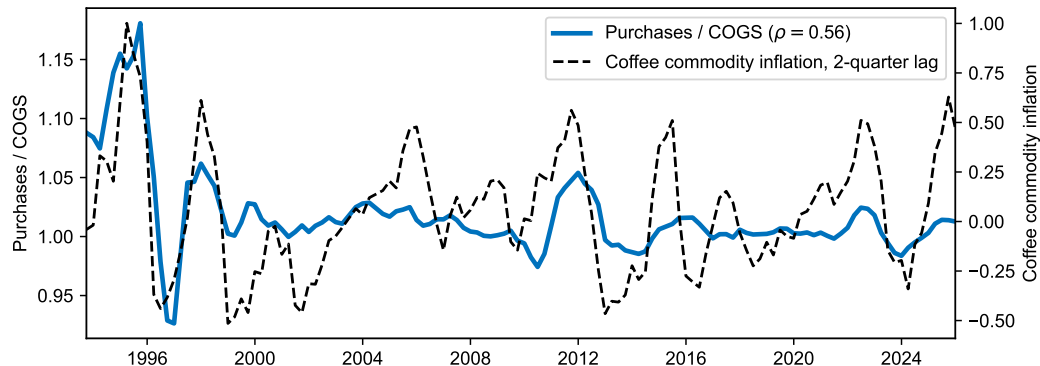
- Log-linearizing:

$$\widehat{\left(\frac{\text{Sales}_t}{\text{COGS}_t}\right)} = \phi \left(\widehat{\left(\frac{\text{Purchases}_t}{\text{COGS}_t}\right)} \right) - \phi (\hat{Y} - \hat{D}).$$

- Absent changes in quantities and markups, ϕ identified by sensitivity of Sales/COGS to Purchases/COGS.

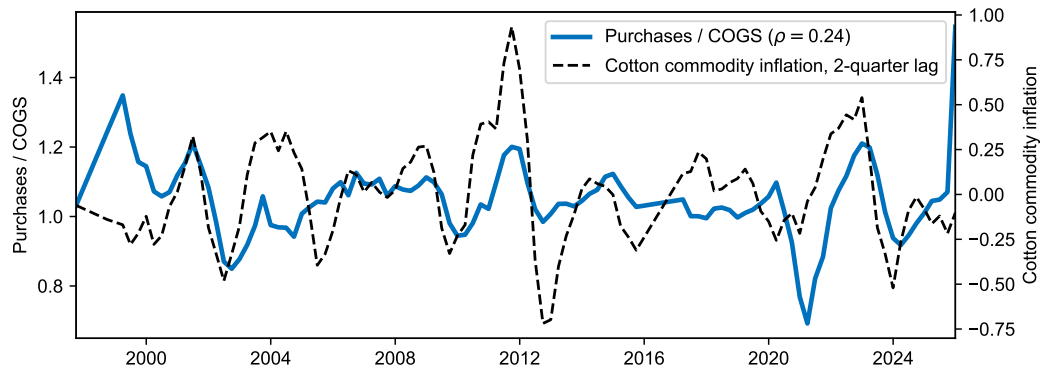
Does Purchases/COGS capture variation in c_t/h_t ? Example: Starbucks

- Use 10Qs for sales, COGS, purchases = $\Delta \text{BookValueInventory} + \text{COGS}$.
- Compare to ICO Arabica coffee bean commodity price.



Does Purchases/COGS capture variation in c_t/h_t ? Example: Gildan

- Use 10Qs for sales, COGS, purchases = $\Delta\text{BookValueInventory} + \text{COGS}$.
- Compare to Cotlook benchmark cotton spot price.



Estimating ϕ across all firms

- Quarterly financial reports for public firms:
 - Exclude firms in information (NAICS 51), finance/insurance/real estate (NAICS 52–53).
 - Observations with strictly positive sales, COGS, and inventory stock.
 - Firms using AAC or FIFO. (Under LIFO, $h_t \approx c_t$.)
 - Change in total assets over past year $< 30\%$ (avoid mergers / spin-offs).
- Specification:

$$\log(\text{Sales}/\text{COGS})_{it} = \phi \log(\text{Purchases}/\text{COGS})_{it} + \alpha_{iq} + \varepsilon_{it}.$$

where α_{iq} are firm $\times$ quarter-of-year FEs.

Estimating ϕ across all firms

- Specification:

$$\log(\text{Sales}/\text{COGS})_{it} = \phi \log(\text{Purchases}/\text{COGS})_{it} + \alpha_{iq} + \varepsilon_{it}.$$

- Change in relative quantities of purchases / shipments $\hat{Y}_t - \hat{D}_t$ is omitted variable:

$$\hat{\phi} \rightarrow \phi \left[1 - \underbrace{\frac{\text{Var}(\hat{Y}_t - \hat{D}_t)}{\text{Var}\left(\widehat{\frac{\text{Purchases}}{\text{COGS}}}\right)}}_{\text{Attenuation bias} < 0} + \underbrace{\frac{-\text{Cov}(\hat{c}_t - \hat{h}_t, \hat{Y}_t - \hat{D}_t)}{\text{Var}\left(\widehat{\frac{\text{Purchases}}{\text{COGS}}}\right)}}_{\text{Forward-buying bias} > 0} \right].$$

- Seasonal variation in $\hat{Y}_t - \hat{D}_t$: firm $\times$ quarter-of-year FEs.
- Attenuation bias: IV using NAICS-4 industry $\times$ Känzig (2021) oil shocks.
- IV estimates are upper bound if firms engage in forward buying ($\hat{Y}_t \downarrow$ when $\hat{c}_t \uparrow$).

Estimating ϕ : All firms

- OLS: $\phi \in [0.12, 0.15]$.
- NAICS-4 $\times$ Känzig (2021) oil shock IV: $\phi \in [0.35, 0.4]$.
- Extent of accounting cost pricing $(1 - \phi) \approx 60\text{--}80\%$.

	<i>Log(Sales / COGS)</i>			
	Levels		First Differences	
	OLS (1)	IV (2)	OLS (3)	IV (4)
Log(Purchases / COGS)	0.147** (0.015)	0.380** (0.104)	0.126** (0.014)	0.351** (0.079)
Firm $\times$ Quarter-of-Year FEs	Yes	Yes	Yes	Yes
<i>N</i>	382 211	369 469	345 468	333 935
<i>R</i> ²	0.86	0.86	0.09	0.04

Estimating ϕ : Comparative statics

① Homogeneous outputs or observable costs.

- Rauch (1999): Industries with reference prices (r) or organized exchanges (w) vs. administered prices (n).

② Competitors pricing off replacement costs.

- Share of firms in industry using LIFO in industry from Economic Census.

③ Inventory lags.

- h_t moves more slowly toward c_t for firms with lower inventory turnover.
- Firm inventory turnover, $\delta = \text{COGS}/\text{Inventories}$.

Replacement cost pricing in commodity industries

- $\uparrow \phi$ with reference prices / organized exchanges, or where competitors use LIFO.

	Levels		<i>Log(Sales / COGS)</i>			
			First Differences			
	OLS (1)	OLS (2)	OLS (3)	OLS (4)	IV (5)	IV (6)
Log(Purchases / COGS)	0.122** (0.016)	0.125** (0.019)	0.101** (0.017)	0.103** (0.016)	-0.268 (0.376)	-0.136 (0.265)
× Reference prices (<i>r</i>)	0.079** (0.029)		0.074* (0.042)		1.006* (0.599)	
× Organized exchange (<i>w</i>)	0.254** (0.062)		0.168** (0.051)		0.720** (0.329)	
× Industry LIFO share		0.185** (0.090)		0.144** (0.068)		0.744** (0.353)
Firm × Quarter-of-Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	251 379	276 804	227 964	251 033	225 546	242 882
<i>R</i> ²	0.88	0.88	0.09	0.09	0.08	0.01

More weight on recent costs when $\uparrow$ turnover

- More weight on recent replacement costs when inventory turns over faster ($\uparrow \delta$).

	$\text{Log}(\text{Sales}_{t \rightarrow t+4} / \text{COGS}_{t \rightarrow t+4})$		$\text{Log}(\text{Sales}_{t \rightarrow t+4} / \text{COGS}_{t \rightarrow t+4})$	
	Levels		First Differences	
	(1)	(2)	(3)	(4)
$\text{Log}(\text{Purchases}_{t \rightarrow t+4} / \text{COGS}_{t \rightarrow t+4})$	0.070 (0.070)	0.062 (0.071)	0.086** (0.024)	0.083** (0.024)
$\text{Log}(\text{Purchases}_{t \rightarrow t+4} / \text{COGS}_{t \rightarrow t+4}) \times \log(\delta)$	0.183** (0.064)		0.065** (0.027)	
$\text{Log}(\text{Purchases}_{t \rightarrow t+4} / \text{COGS}_{t \rightarrow t+4}) \times \log(\delta) \times \text{FIFO}$		0.088 (0.061)		0.043* (0.026)
$\text{Log}(\text{Purchases}_{t \rightarrow t+4} / \text{COGS}_{t \rightarrow t+4}) \times \log(\delta) \times \text{AAC}$		0.313** (0.081)		0.099** (0.033)
Firm $\times$ Quarter-of-Year FEs	Yes	Yes	Yes	Yes
N	365 785	365 785	338 437	338 437
R^2	0.89	0.89	0.11	0.11

Summary of estimates

- Average extent of accounting cost pricing $(1 - \phi) \approx 60\text{--}80\%$.
 - Accounting cost pricing $\geq 85\%$ for firms in industries with differentiated products (n).
 - Replacement cost pricing in industries with reference prices (r) or exchanges (w).
- Dependence on historical costs not due to inattention to replacement costs.
 - Our test uses purchases already made by firm!
 - Inventory turnover matters.

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Why price off accounting costs?

- “The need for objective and systematic measurement forces the cost standard to be backward-looking—or at least sideward- rather than forward-looking. [...] Any attempt to forecast input costs or to put them on a current replacement basis, which necessarily entails a forecast, would add to expenses, introduce a subjective element into cost calculations, and complicate the task of managerial control. Bygones cannot be mere bygones if they are the sole sources of systematic information.”

— Okun (1981), *Prices and Quantities*

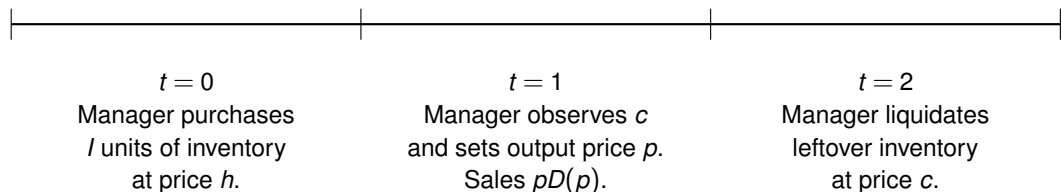
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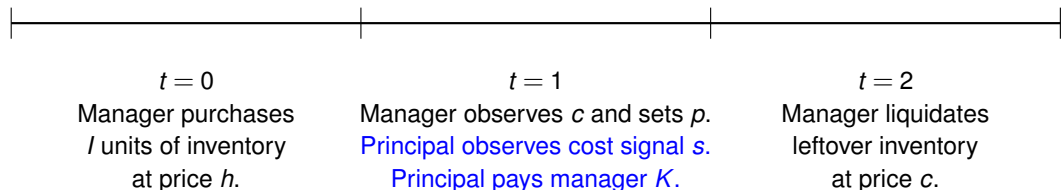
- ① Ingredient #1: Replacement cost unverifiable, but accounting metrics observable.
- ② Ingredient #2: Misaligned incentives between principal and manager.

Two-period model: Frictionless



- No discounting. Assume I fixed and $D(p) < I$ for all possible p (no stockouts).
- $V^{\text{FB}} = \max_p \{-hl + pD(p) + c(I - D(p))\}$.
- $p^* = \arg \max_p (p - c)D(p)$.

Two-period model: Frictions



- Principal delegates firm operations to manager.
- **Friction 1:** Replacement cost c is manager's private information.
- Public signal $s \sim N(c/h, \tau^{-1})$. Friction disappears as precision $\tau \rightarrow \infty$.

Two-period model: Frictions

- **Friction 2:** Misaligned incentives.

$$V^{\text{manager}} = \max_p \left\{ \underbrace{-hl + pD(p) + c(l - D(p))}_{\text{True firm value}} + \underbrace{(\alpha - 1)p}_{\text{Pricing wedge}} + \underbrace{K(pD(p), hD(p); s)}_{\text{Incentive payment}} \right\}.$$

- When $\alpha \neq 1$, manager wants to set price lower or higher than optimal.
E.g., Baumol (1959), Bertrand and Mullainathan (2003), Graham et al. (2005).
- Manager preference α is private information.
- Principal can **incentivize manager**, but can only condition payment on observed info.

Two-period model: Frictions

- Why can't principal condition payment on liquidation proceeds at $t = 2$?
 - Liquidation proceeds reveal c to both parties.
 - We assume that contract must be tied to performance observable at $t = 1$
 $\Rightarrow$ need to monitor and incentivize manager in a timely manner.
- Why can't manager just tell principal c ?
 - Manager always has incentive to misreport c in order to justify desired price.
- Principal's trade-off:
 - Stronger incentives limit extent to which manager preference α affects price...
 - ...but also lead manager to disregard valuable info about c .

Two-period model

- Log-linearize around $h = c = 1$ and $\alpha = 1$. Let $\hat{x}$ be log-deviation of x .
- Define $\sigma_\alpha^2 = \mathbb{E}[\hat{\alpha}^2]$ and $\sigma_\varepsilon^2 = \mathbb{E}[(\hat{c} - \hat{h})^2]$.
- Local demand elasticity η . I.e., $\hat{D} = -\eta \hat{p}$.
- Assume principal's payment quadratic around target accounting markup $\mu(s)$,

$$K(pD(p), hD(p); s) = -\frac{\kappa}{2} \left(\frac{pD(p)}{hD(p)} - \mu(s) \right)^2. \quad (1)$$

Pricing without incentives

Proposition (Pricing without incentives)

Suppose there is no incentive payment ($\kappa = 0$). Then, the price set by manager is

$$\hat{p} = \hat{c} + \psi_0 \hat{\alpha}, \quad \text{where} \quad \psi_0 = \frac{1}{\eta - 1} \left(\frac{\eta}{\eta - 1} \right)^\eta.$$

- ψ_0 = elasticity of price to manager preference α when $\kappa = 0$.
 - Decreasing in demand elasticity η .
 - Managerial slack harder to sustain in competitive industries.
Hart (1983), Giroud and Mueller (2010, 2011).

Optimal contract parameters

Proposition (Principal's optimal contract)

Given quadratic incentive payment (1), the principal's optimal choice of $\mu(s)$ and κ are

$$\underbrace{\mu^*(s) = \frac{\eta}{\eta - 1} \exp\left(\frac{\tau \sigma_\varepsilon^2}{1 + \tau \sigma_\varepsilon^2} s\right)}_{\text{Target accounting markup}}, \quad \underbrace{\kappa^* = \frac{1}{\eta} \left(\frac{\eta}{\eta - 1}\right)^\eta \frac{\sigma_\alpha^2}{\sigma_\varepsilon^2} (1 + \tau \sigma_\varepsilon^2)}_{\text{Strength of incentives}}.$$

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- Principal “marks-to-market” inventory using signal s .

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- Principal “marks-to-market” inventory using signal s .
- Stronger incentives κ^* when...
 - Risk of discretion $\gg$ value of private info, $\sigma_\alpha^2 / \sigma_\varepsilon^2$.
 - Demand elasticity η low.
 - Cost observed precisely ($\tau \uparrow$).

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Pricing under optimal contract

Proposition (Pricing with incentives)

Under payment's optimal contract, manager sets price

$$\hat{p} = \underbrace{\phi [\hat{c} + \psi \hat{\alpha}]}_{\text{Replacement cost pricing}} + \underbrace{(1 - \phi) \hat{h}}_{\text{Accounting cost pricing}},$$

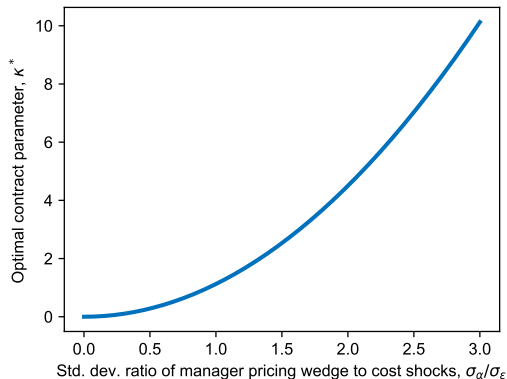
where

$$1 - \phi = \frac{\sigma_{\alpha}^2 / \sigma_{\varepsilon}^2}{\sigma_{\alpha}^2 / \sigma_{\varepsilon}^2 + \psi_0^{-2} + \tau \sigma_{\alpha}^2}, \quad \text{and} \quad \psi = \frac{1}{1 + \tau \sigma_{\alpha}^2 \psi_0^2} \psi_0.$$

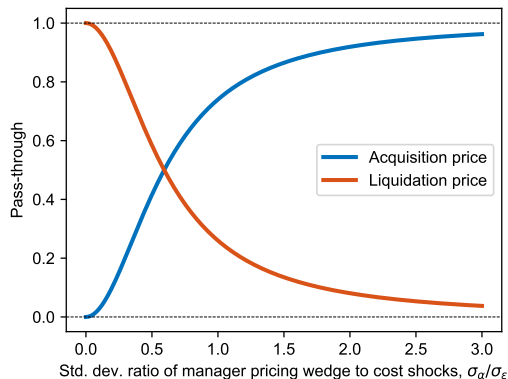
- Firm sets blended average of **replacement cost** and **accounting cost** price.
- $1 - \phi$ is the extent of accounting cost pricing.
 - Increases with risk of managerial discretion ($\uparrow \sigma_{\alpha}^2 / \sigma_{\varepsilon}^2$).
 - Falls with competition ($\uparrow \eta$) and cost observability ($\uparrow \tau$).

Pricing under optimal contract

- As risk of manager discretion grows, κ^* rises and ϕ falls.

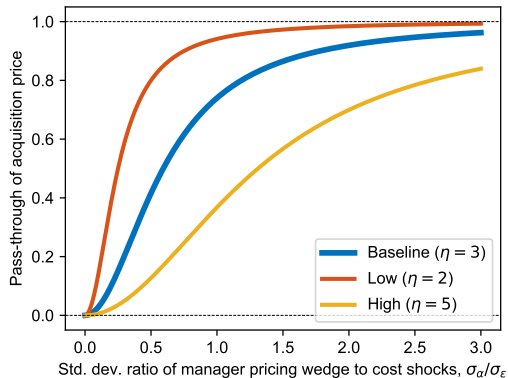


(a) Optimal contract parameter κ^* .

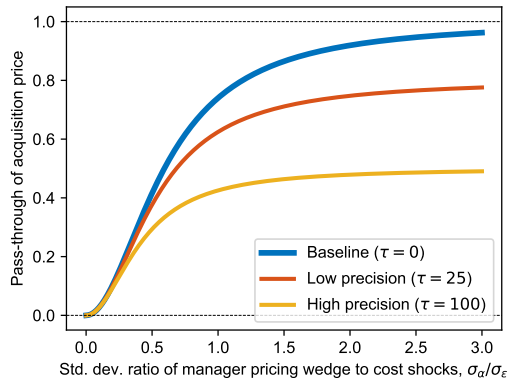


(b) Replacement vs. accounting cost pricing.

Comparative statics



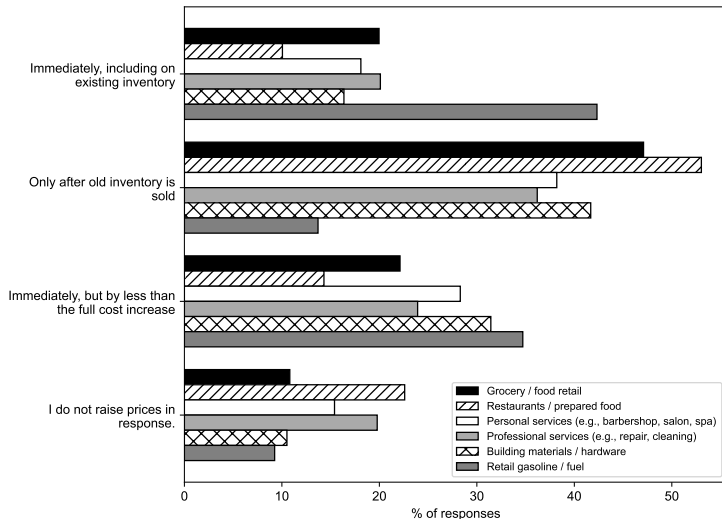
(a) Demand elasticity η .



(b) Signal precision τ .

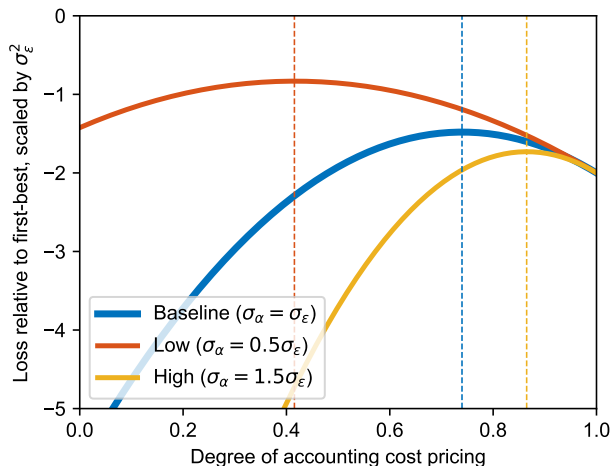
- More replacement cost pricing when costs observable or competition high.

Comparative statics



- Exception: gas stations.
- “The exceptions that occur to me arise when processed raw materials are sold on auction markets, as in the case of flour or soybean meal.” (Okun 1981).
- (Anecdotal) exception: sellers of gold/silver jewelry.

Losses relative to first best



- With optimal contract and $\tau = 0$,
$$\mathcal{L} = \mathbb{E}[V^{\text{FB}} - V] \approx (\eta - 1)(1 - \phi)\sigma_\varepsilon^2.$$
- If $\sigma_\varepsilon^2 = 0$, $\mathcal{L} = 0$.
- If $\sigma_\alpha^2 = 0$, $\phi = 1$ and $\mathcal{L} = 0$.
- $\mathcal{L}$ nonmonotonic in η .
($\uparrow$ losses but $\phi \rightarrow 1$.)

Other pricing puzzles

- Main result: Unverifiable info and principal–agent frictions mean:
 - Prices incorporate info irrelevant to p^{FB} (e.g., historical input prices).
 - Prices neglect info germane to p^{FB} but unverifiable (e.g., replacement costs).

Other pricing puzzles

- Main result: Unverifiable info and principal–agent frictions mean:
 - Prices incorporate info irrelevant to p^{FB} (e.g., historical input prices).
 - Prices neglect info germane to p^{FB} but unverifiable (e.g., replacement costs).
- Fixed costs: Suppose both c and overhead cost F are stochastic.
 - Pass through innovations in F with same weight as $\hat{h}$, $(1 - \phi)$.
 - Long history of survey evidence. Hall and Hitch (1939), Altomonte et al. (2015), Bewley (2025).
- Demand shocks: Suppose demand elasticity η is stochastic.
 - Pass through innovations in optimal markup with same weight as $\hat{c}$, ϕ .
 - Empirical evidence. Cagan (1979), Bils and Chang (2000), Fabiani et al. (2006), Gagnon and Lopez-Salido (2020), Kohler et al. (2026).

Alternative explanations

- **Earnings smoothing.** Graham et al. (2005), Terry (2023).
 - Earnings depend on quantities $\Rightarrow$ depends on what other firms do.
 - If quantities don't change in equilibrium (e.g., all firms set same price and industry demand inelastic), then earnings smoothing = setting target accounting markup.
- **Fairness.** Kahneman et al. (1986), Rotemberg (2005, 2011), Eyster et al. (2021).
 - Similar structure: outsider tries to evaluate whether price is justified.
 - Fairness binds more on price increases than decreases. But estimates of ϕ symmetric.

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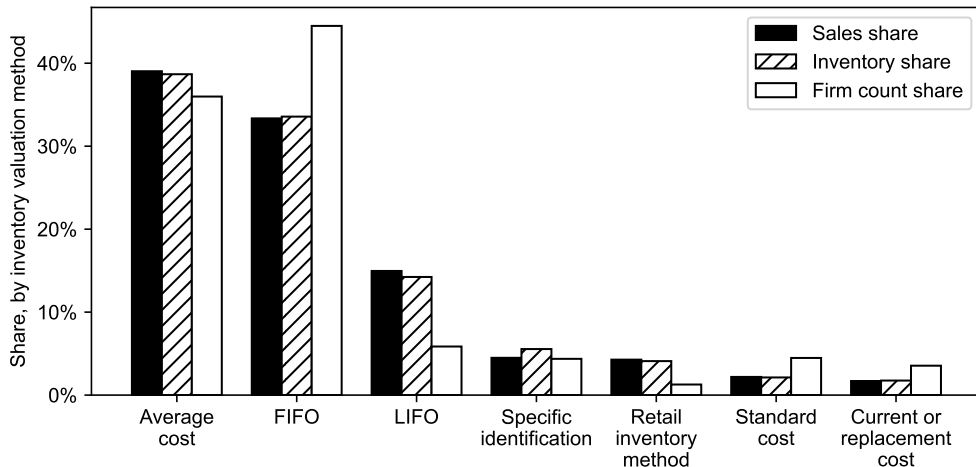
Macro Rigidity

Calculating accounting costs

- In two-period setting, book value of inventory is unambiguous (hl).
- In dynamic setting, depends on how firm apportions cost across past purchases. E.g.,
 - Average acquisition cost (AAC).
 - First-in-first-out (FIFO).
 - Last-in-first-out (LIFO).
- Under all three, COGS reflects historical purchase costs. Methods differ in the weight they place on recent vs. more distant periods.
- $\Rightarrow$ Connection between accounting cost pricing and canonical sticky-price models.

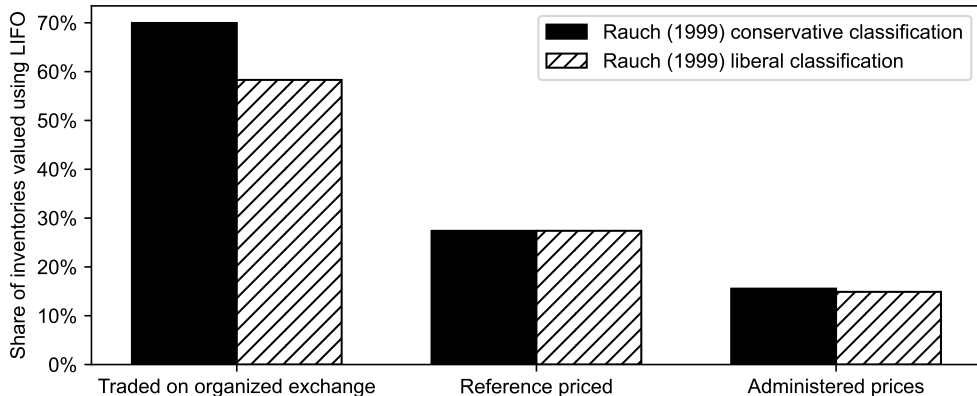
AAC and FIFO most common inventory valuation methods

- Valuation methods used by inventory-carrying public firms over 2010–2025:



AAC/FIFO common when accounting cost pricing likely

- LIFO concentrated in industries with homogeneous outputs (organized exchanges) or inputs (reference-priced).



Price dynamics under average acquisition cost (AAC)

- Average acquisition cost (AAC) accounting:

$$h_t l_t = (l_{t-1} - D_t) h_{t-1} + Y_t c_t.$$

$\Rightarrow h_t$ is average of h_{t-1} and c_t , weighted by leftover inventory vs. new purchases.

- Log-linearizing around steady state with **shipment-inventory ratio** $\delta = D^{ss}/I^{ss}$,

$$\hat{p}_t = \hat{h}_t = (1 - \delta) \hat{h}_{t-1} + \delta \hat{c}_t.$$

- Recursive formulation for prices looks familiar...!

Price dynamics under average acquisition cost (AAC) vs. Calvo

- Given a path of cost deviations from steady-state $\{\hat{c}_t\}_{t=-\infty}^{\infty}$,

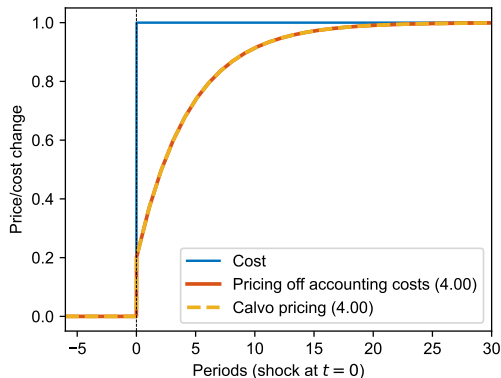
$$\hat{p}_t^{\text{AAC}} = \delta \sum_{k=0}^{\infty} (1 - \delta)^k \hat{c}_{t-k},$$

$$\hat{p}_t^{\text{Calvo}} = \theta \sum_{k=0}^{\infty} (1 - \theta)^k \hat{c}_{t-k} + \underbrace{\theta [1 - \beta (1 - \theta)] \sum_{k=0}^{\infty} \sum_{s=0}^{\infty} \beta^s (1 - \theta)^{k+s} \mathbb{E}_{t-k} [\hat{c}_{t+s-k} - \hat{c}_{t-k}]}_{\text{Due to forward-looking reset prices}},$$

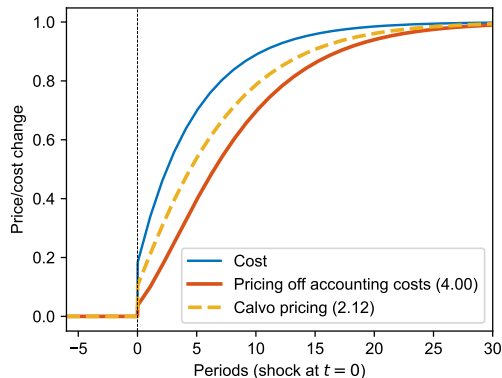
- AAC: δ is steady-state shipment-inventory ratio.
- Calvo: θ is fraction of firms able to reset prices each period, and β is discount factor.
- Paths coincide if $\delta = \theta$ and either $\beta = 0$ or $\mathbb{E}_{t+s}[\hat{c}_{t+s}] = \hat{c}_t$ for all t, s .
- Behaves like “myopic Calvo” (Minton and Wheaton 2022), but not due to inattention.

#1: More delayed adjustment to slow-moving shocks

- Set $\delta = \theta = 0.2$. Shock path, unexpected before $t = 0$: $\hat{c}_t = 1 - \rho^{t+1}$.



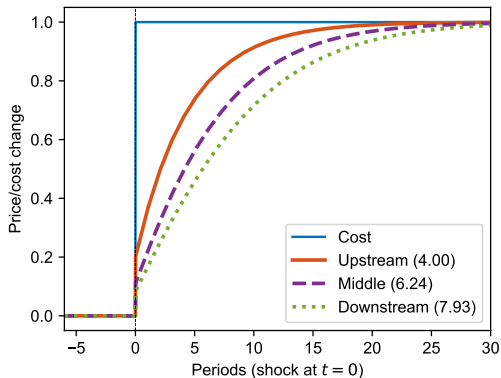
(a) Immediate shock ($\rho = 0$).



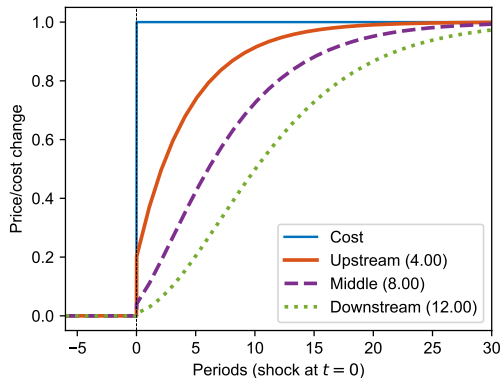
(b) Slow-moving shock ($\rho = e^{-0.2}$).

#2: Amplification of delays in supply chains

- Each firm turns cost shock into slower-moving shock for downstream firm.

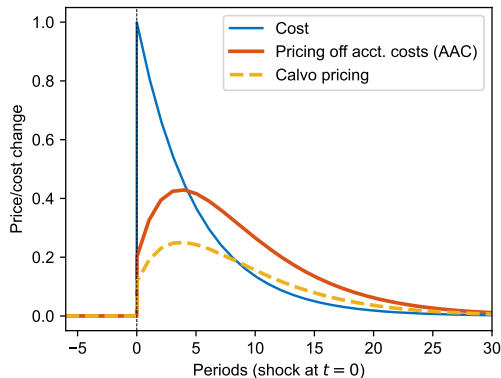


(a) Calvo.

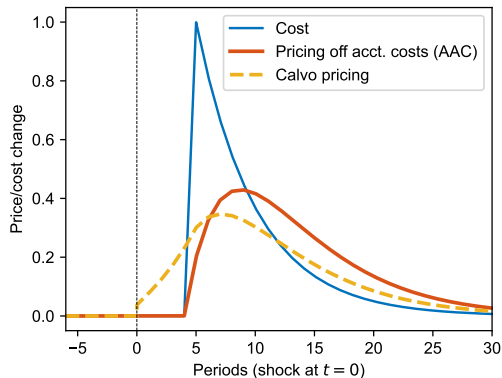


(b) Pricing off accounting costs (AAC).

#3: Increased responsiveness to transitory shocks



(a) Unanticipated cost increase at $t = 0$.



(b) News at $t = 0$ about cost increase at $t = 5$.

- More volatility and less persistence of inflation. *Bils and Klenow (2004), Bils et al. (2012).*

#4: Forward buying delays pass-through of cost increases

- First-order approximation for prices did not require keeping track of inventory stocks:

$$\hat{p} = (1 - \delta)\hat{h}_{t-1} + \delta\hat{c}_t.$$

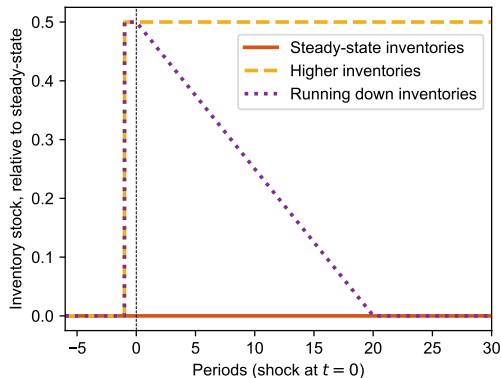
- Effects of stocking up on inventories show up at second-order:

$$\hat{h}_t = \underbrace{(1 - \delta)\hat{h}_{t-1} + \delta\hat{c}_t}_{\text{First order}} + \underbrace{\frac{1}{2}\delta(1 - \delta)(\hat{c}_t - \hat{h}_{t-1})^2}_{\text{Second-order correction}} + \underbrace{\delta(\hat{c}_t - \hat{h}_{t-1})(\hat{y}_t - \hat{l}_t)}_{\text{Effect of inventories / purchases}}.$$

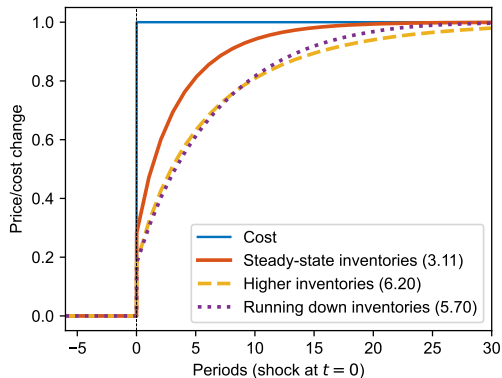
- If purchases low relative to inventory stock when replacement costs rise $\Rightarrow$ lower price.

#4: Forward buying delays pass-through of cost increases

- Illustrate with exogenous inventory paths in second-order approximation:

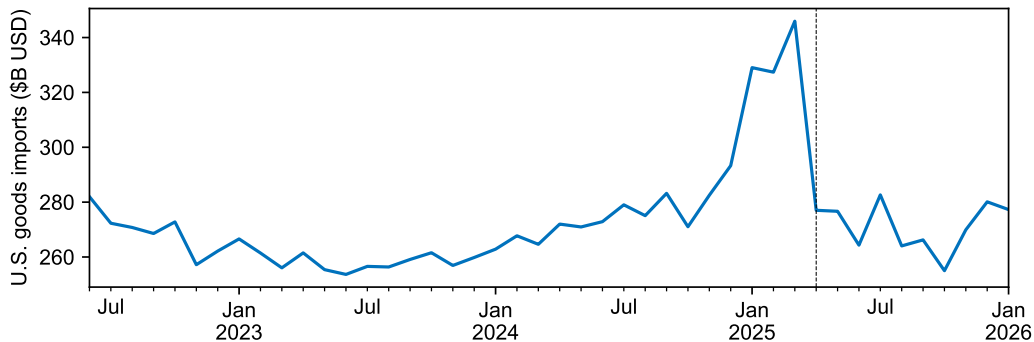


(a) Inventory paths.



(b) Pricing off accounting costs (AAC).

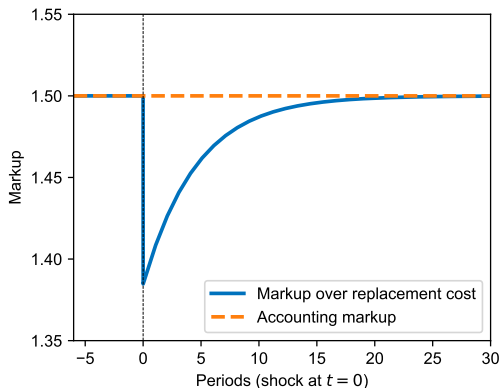
#4: Forward buying delays pass-through of cost increases



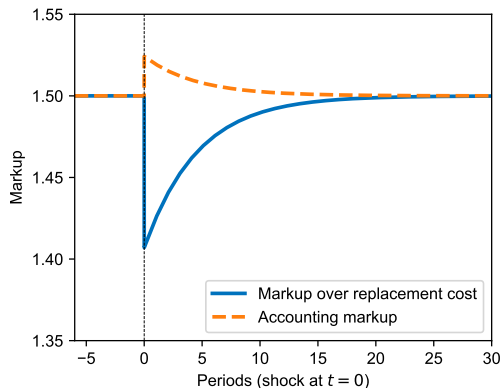
- Sticky prices: Need “myopic expectations” to amplify delays. Minton and Wheaton (2022).
- But firms clearly stocking up in anticipation! Not myopic.
- Accounting cost pricing: Anticipation **delays**, rather than speeds up, pass-through.

#5: Response of measured vs. true markups

- Measured markups can be acyclical/procyclical even if prices fall vs. replacement cost.
- Example: Steady-state markup of 1.5, with 10% permanent shock to costs.



(a) Full accounting cost pricing ($\phi = 0$).



(b) Partial accounting cost pricing ($\phi = 0.2$).

FIFO and LIFO

- Given a path of cost deviations from steady-state $\{\hat{c}_t\}_{t=-\infty}^{\infty}$,

$$\hat{p}_t^{\text{FIFO}} = \hat{c}_{t-1}/\delta,$$

$$\hat{p}_t^{\text{LIFO}} = \hat{c}_t,$$

FIFO and LIFO

- Given a path of cost deviations from steady-state $\{\hat{c}_t\}_{t=-\infty}^{\infty}$,

$$\hat{p}_t^{\text{FIFO}} = \hat{c}_{t-1/\delta},$$

$$\hat{p}_t^{\text{LIFO}} = \hat{c}_t,$$

$$\hat{p}_t^{\text{LIFO-}\delta} = \delta \sum_{k=0}^{1/\delta-1} \hat{c}_{t-k},$$

[Firms buy inventory every $1/\delta$ periods.]

FIFO and LIFO

- Given a path of cost deviations from steady-state $\{\hat{c}_t\}_{t=-\infty}^{\infty}$,

$$\hat{p}_t^{\text{FIFO}} = \hat{c}_{t-1/\delta},$$

$$\hat{p}_t^{\text{LIFO}} = \hat{c}_t,$$

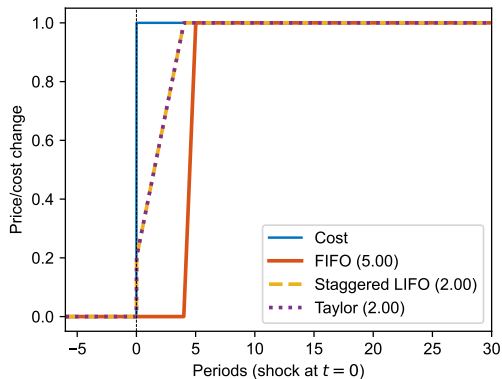
$$\hat{p}_t^{\text{LIFO-}\delta} = \delta \sum_{k=0}^{1/\delta-1} \hat{c}_{t-k}, \quad \text{[Firms buy inventory every } 1/\delta \text{ periods.]}$$

$$\hat{p}_t^{\text{Taylor}} = \frac{1}{N} \sum_{k=0}^{N-1} \hat{c}_{t-k} + \underbrace{\frac{1}{N} \frac{1-\beta}{1-\beta^N} \sum_{k=0}^{N-1} \sum_{s=0}^{N-1} \beta^s \mathbb{E}_{t-k} [\hat{c}_{t+s-k} - \hat{c}_{t-k}]}_{\text{Due to forward-looking reset prices}}.$$

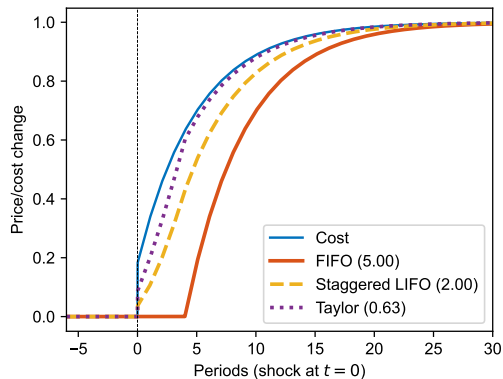
- LIFO w/ intermittent purchases (LIFO- δ): Firms stock up on inventory every $1/\delta$ periods.
- Taylor: Firm resets price every N periods, and β is discount factor.
- LIFO- δ and Taylor coincide if $\delta = 1/N$ and either $\beta = 0$ or $\mathbb{E}_{t+s}[\hat{c}_{t+s}] = \hat{c}_t$ for all t, s .

FIFO, LIFO, and Taylor pricing

- Similar intuitions as AAC vs. Calvo.



(a) Immediate shock.



(b) Slow-moving shock.

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Estimating the Extent of Accounting Cost Pricing

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Macro Rigidity

Accounting cost pricing $\Rightarrow$ macro rigidity?

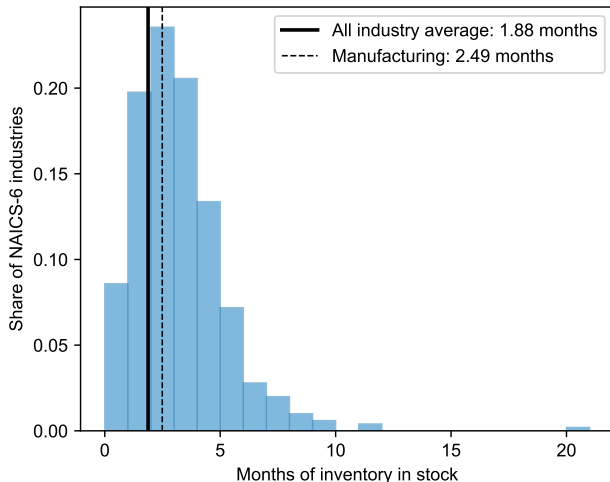
- “Industry experts estimate it takes at least nine months for raw bean prices to filter through to coffee drinkers [...] because U.S. roasters typically hold about two to three months’ worth of bean stocks on average and need another two to three months to roast and package their products.”
— *Reuters*, Dec 19 2025
- “Cocoa—the key ingredient in chocolate—has been steadily falling in price from its all-time high in 2024. [...] But that drop hasn’t trickled down to what consumers are paying. [...] The reason: The Easter chocolate on shelves today was produced using cocoa purchased at its extreme high.”
— *CNN Business*, Apr 5 2026
- “It is reasonably predictable that today’s changes in cattle prices will show up on the supermarket counter only after a lag; the butcher does not keep changing price tags in pace with changes in some putative replacement cost.”
— Okun (1981), *Prices and Quantities*

Quantitative input–output model of U.S. economy

Variable	Description	Source
Ω	Input–output matrix	BEA 2017 405-industry input-output table
δ_i	Shipment–inventory ratios	Economic Census 2017 COGS / Inventories

- Inventories for mining (21), manufacturing (31–33), wholesale (42) & retail (44–45).
 - Underestimate if firms use off-balance-sheet instruments (e.g., forwards) to lock in prices.
- Problem: BEA treats transportation, wholesale, and retail trade as “margin industries.”
 - Reverse engineer flows of goods through trade industries using BEA Margins table.
 - “Trade-adjusted” input–output network with 843 industries.

Inventory–shipment ratios from Economic Census

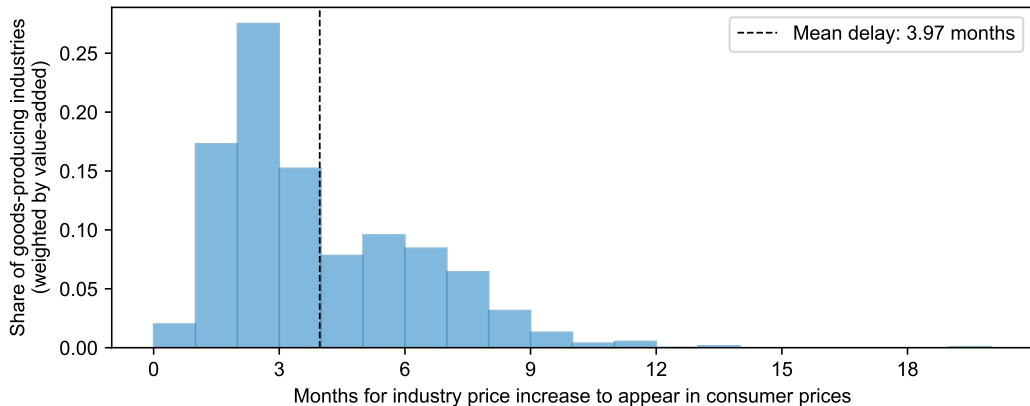


Industries with largest inventory–shipment ratios:

- Wineries.
- Biological product mfg.
- Aircraft mfg.
- Distilleries.
- Aerospace products and parts mfg.

3–4 month delay between industry price changes and CPI

- Change in price of goods-producing industry takes 4.0 months on average to show up in consumer prices under FIFO (3.2 months under AAC).

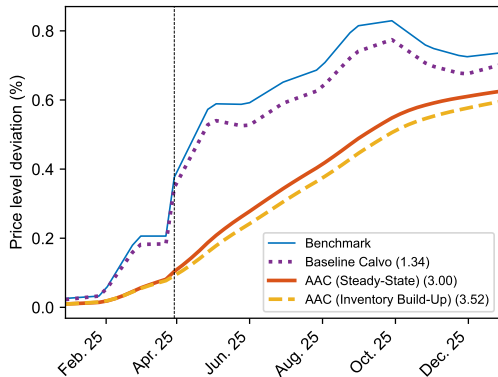


Effect of 2025 tariffs on consumer price index

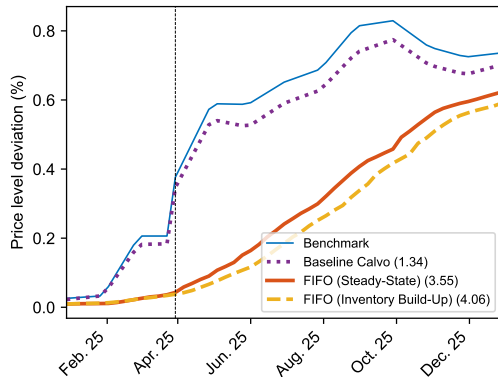
- Monthly effective tariff rates by commodity from Jan '25 to Jan '26.
- Simulate how shocks passed through input–output network (AAC or FIFO).
- Two assumptions about inventories:
 - Steady-state inventories.
 - Stockpiling given observed import quantities (assume importing industry holds excess).
- Also compare to benchmark Calvo model, calibrated á la Rubbo (2023).
 - Standard treatment: Trade industry rigidity only applies to trade margins.
 - Industry-specific price rigidities from Pasten et al. (2020), $\beta = (0.99)^{1/12}$.

Effect of 2025 tariffs on consumer price index

- Accounting cost pricing creates delays relative to flex-price benchmark (blue).
- Calvo (á la Rubbo 2023) adjusts too quickly, largely due to lack of trade industries.



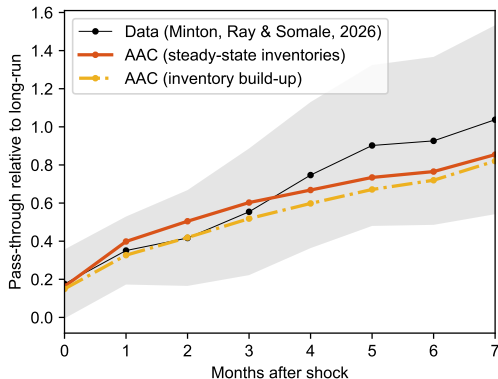
(a) Average acquisition cost (AAC).



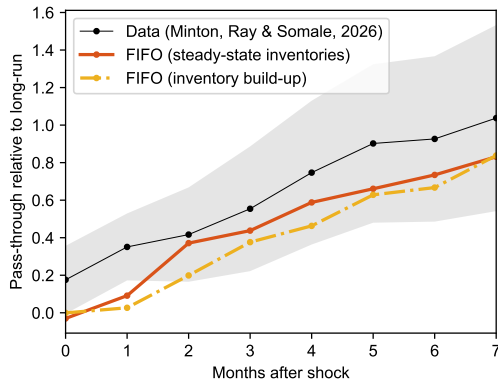
(b) First-in-first-out (FIFO).

Effect of 2025 tariffs on consumer price index: Pass-through delays

- Pass-through from distributed lag regression: $\Delta p_{it} = \sum_{k=0}^K \beta_k \Delta p_{it-k}^{LR} + \alpha_i + \lambda_t + \varepsilon_{it}$.
- Use PCE category variation (with time & category FEs), as in Minton et al. (2026).



(a) Average acquisition cost (AAC).



(b) First-in-first-out (FIFO).

Effect of 2025 tariffs on consumer price index: Model vs. data

- Accounting cost models appear to capture price responses in data.
- Outperform off-the-shelf Calvo model (largely due to trade industries).

	<i>PCE category monthly inflation: Data</i>				
	(1)	(2)	(3)	(4)	(5)
Inflation in AAC Model	1.331** (0.429)			1.349** (0.420)	
Inflation in FIFO Model		1.374** (0.418)			1.384** (0.418)
Inflation in Calvo Model			-1.027 (3.861)	-2.251 (3.568)	-2.295 (3.609)
Date FEs	Yes	Yes	Yes	Yes	Yes
PCE Category FEs	Yes	Yes	Yes	Yes	Yes
<i>N</i>	4876	4876	4876	4876	4876
<i>R</i> ²	0.04	0.04	0.03	0.04	0.04

Conclusion

- Widespread reports that firms set prices using past inventory purchase costs.
- In data: firms put 60–80% weight on accounting costs, $\geq 85\%$ if administered prices.
- We develop a model of [accounting cost pricing](#).
- Synthesizes other pricing puzzles:
 - Use irrelevant info: Hedging gains/losses, changes in fixed costs.
 - Neglect relevant info: Demand shocks, replacement costs, anticipated future costs.
- Can resemble sticky-price models, generate substantial delays in macro adjustment.

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Extra Slides

Survey Evidence

Quantitative Exercise

The Standard Theory: Firm Problem

- Given inventory I_{t-1} and current replacement cost c_t , firm value is:

$$V(I_{t-1}, c_t) = \max_{p_t, Y_t} \{p_t D_t - c_t Y_t + \beta \mathbb{E}[V_{t+1}(I_t, c_{t+1})]\},$$

where

$$I_t = I_{t-1} - D_t + Y_t. \quad \text{(Inventory law of motion)}$$

$$mr_t(p_t) = D_t + p_t(dD_t/dp_t). \quad \text{(Marginal revenue)}$$

- FOCs: Set marginal revenue = marginal cost = current replacement cost:

$$mr_t(p_t) = \beta \mathbb{E} \left[\frac{\partial V_{t+1}}{\partial I_t} \right] = c_t.$$

- Some (unsatisfactory) directions: $Y_t \geq 0$, $D_t \leq I_{t-1}$, perishable inventory, convex inventory carrying costs, borrowing constraint.

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Survey Evidence

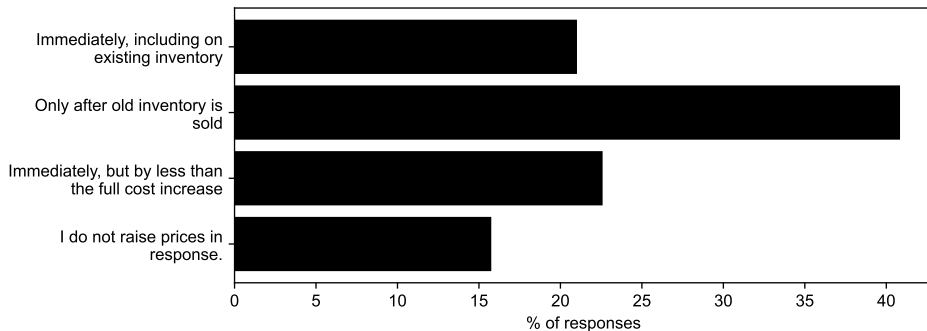
Quantitative Exercise

Survey evidence

- 1 Many firms report waiting to raise prices until inventory at old cost is exhausted.
- 2 Pass-through of changes in overhead costs.
- 3 Limited pass-through of expected future costs.

Survey evidence

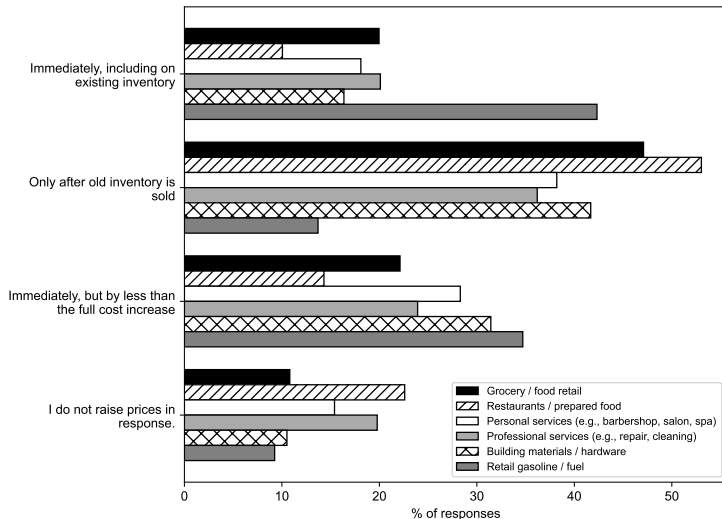
- “Suppose your input cost rises today, but you still have inventory bought at lower cost. When do you raise your sales price?”



Note: Survey of 400 firm price-setters by Schoenle (2026).

- Not about uncertainty about future costs!

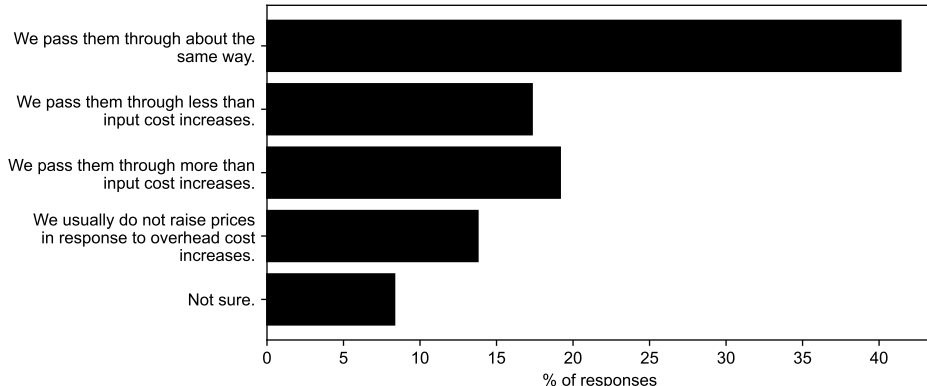
Exceptions



- Exception: gas stations.
- “The exceptions that occur to me arise when processed raw materials are sold on auction markets, as in the case of flour or soybean meal.” (Okun 1981).
- (Anecdotal) exception: sellers of gold/silver jewelry.

Firms' reported pass-through of fixed costs

- “Suppose your overhead costs go up, for example on rent, insurance, salaried staff or equipment leases. How do increases in such overhead costs affect your pricing—compared to increases in per-unit input costs on materials, wholesale goods, shipping per item, or hourly labor per job?”



Prices don't incorporate expected cost changes, but not b/c firms uninformed

- Fed Atlanta BIE survey: Firms' price changes incorporate realized, past cost changes but not expectations of future cost changes.
- Not because firms inattentive: Expectations predict future cost changes.

	<i>Realized Price Change_t</i>		<i>Realized Cost Change_{t+12}</i>	
	(1)	(2)	(3)	(4)
$\mathbb{E}_t[\text{Cost Change}_{t+12}]$	-0.082 (0.204)	-0.169 (0.182)	0.354** (0.090)	0.294** (0.083)
Realized Cost Change _t	1.272** (0.305)	0.979** (0.233)	-0.050 (0.055)	-0.040 (0.048)
Firm FEs	Yes	Yes	Yes	Yes
Date FEs	Yes		Yes	
Date-Sector FEs		Yes		Yes
N	3493	3480	1415	1410
R^2	0.48	0.56	0.65	0.73

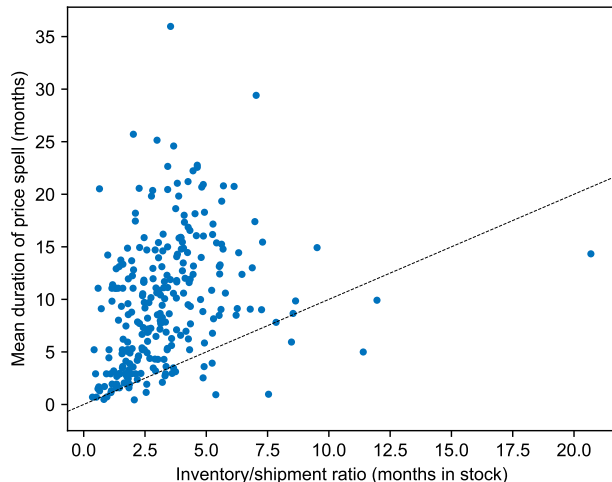
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Survey Evidence

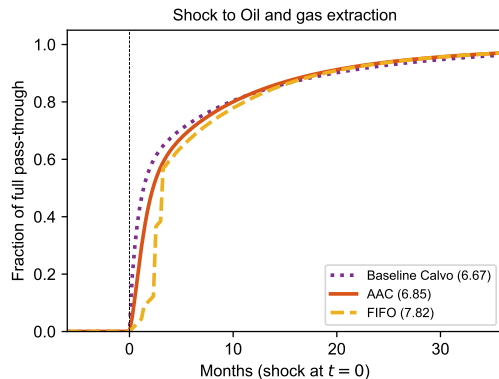
Quantitative Exercise

Inventory–shipment ratios from Economic Census

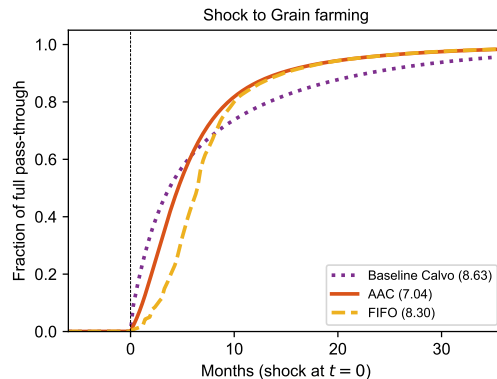


- For most industries, $1/\delta_i < 1/\theta_i$ from Pasten et al. (2020).
- Two factors help accounting cost pricing.
 - 1. Forward-looking reset prices under Calvo.
 - 2. Reroute I–O table to account for margin industries.

Effects of commodity industry shocks on CPI



(a) Oil and gas extraction.



(b) Grain farming.

- Measure cumulative pass-through delay, $\sum_{t=0}^{\infty} (1 - \hat{p}_{CPI,t} / \hat{p}_{CPI,\infty})$.

Effects of commodity industry shocks on CPI

- Despite $1/\delta_i < 1/\theta_i$, accounting cost pricing generates similar delays in price adjustment to benchmark Calvo model.
- AAC or FIFO generate larger cumulative pass-through delay in about 1/2 of cases.

Industry description	Long-run pass-through (%)	Cumulative delay		
		Calvo	AAC	FIFO
Petroleum refineries	4.35	5.70	5.59	6.21
Electric power generation, transmission, and distribution	2.56	7.34	5.74	5.97
Animal (except poultry) slaughtering and processing	1.78	3.41	4.34	4.84
Other basic organic chemical manufacturing	1.53	13.25	10.99	11.82
Beef cattle ranching and farming	1.35	4.75	4.95	5.89
Grain farming	1.27	8.63	7.04	8.30
Plastics material and resin manufacturing	1.02	16.00	11.40	12.38
Iron and steel mills and ferroalloy manufacturing	0.87	18.37	12.10	13.15
Paperboard container manufacturing	0.80	13.55	9.89	10.68
Fruit and tree nut farming	0.79	6.05	5.98	6.91

Effect of 2025 tariffs on consumer price index: Model vs. data

- Using inventory-buildup FIFO and AAC models vs. augmented Calvo model.

	<i>Inflation in PCE data</i>				
	(1)	(2)	(3)	(4)	(5)
Inflation in AAC Model	1.331** (0.429)			0.285 (0.622)	
Inflation in FIFO Model		1.374** (0.418)			1.200* (0.617)
Inflation in Calvo Model			1.895** (0.635)	1.542 (1.123)	0.476 (0.906)
Date FEs	Yes	Yes	Yes	Yes	Yes
PCE Category FEs	Yes	Yes	Yes	Yes	Yes
<i>N</i>	4876	4876	4876	4876	4876
<i>R</i> ²	0.04	0.04	0.04	0.04	0.04

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